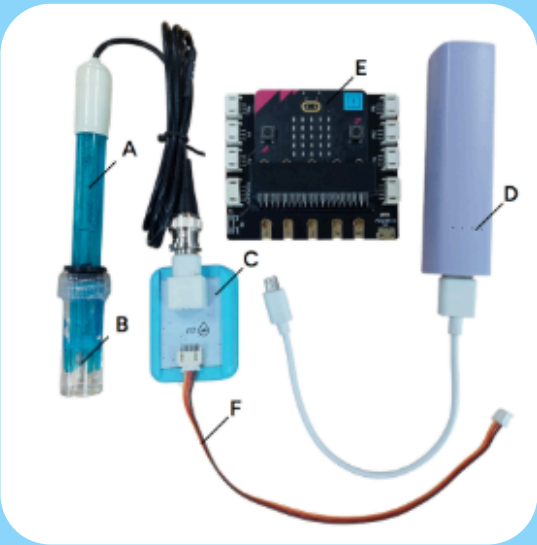


Parts of the water pH meter

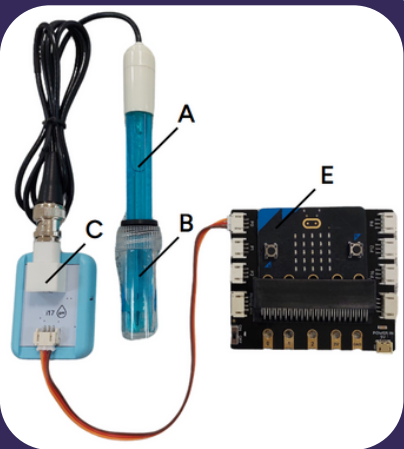


- A - pH Probe
- B - Glass Sensor
- C - pH Sensor Module
- D - Battery Pack
- E - Micro:bit
- F - Connector Cable

How to use a pH Meter

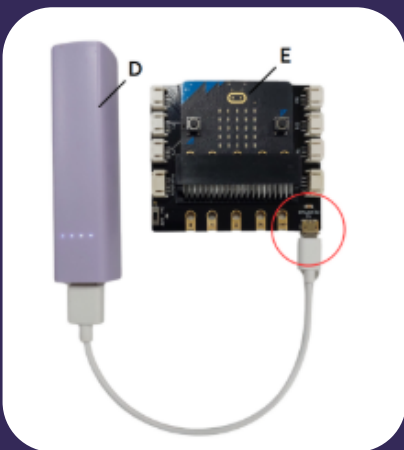
1

To set up the pH meter, connect the pH Sensor Module (C) to port P0 of the Microbit (E) using the Connector Cable (F).



2

Connect the Microbit (E) to a power source such as a battery (G).



3

Collect your water samples using bulb pipettes and/or the BIG blue-capped tubes. Each tube is labelled with a number (1-7), record the sample (eg. water 1) and it's location on your Data Collection Sheet.

Sample Type	Location
Water 1	Royal Park Pond (near the public toilets)

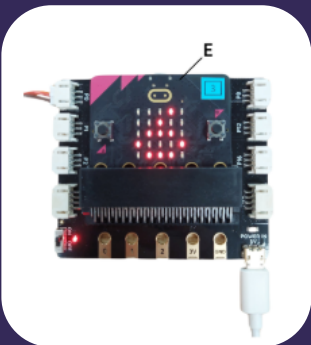
4

To use the pH meter, you need to first unscrew the cap of the Glass sensor (B).



5

Gently dip the Glass sensor (B) into your sample. The LED lights on the Microbit (E) will show you the pH value of the sample.

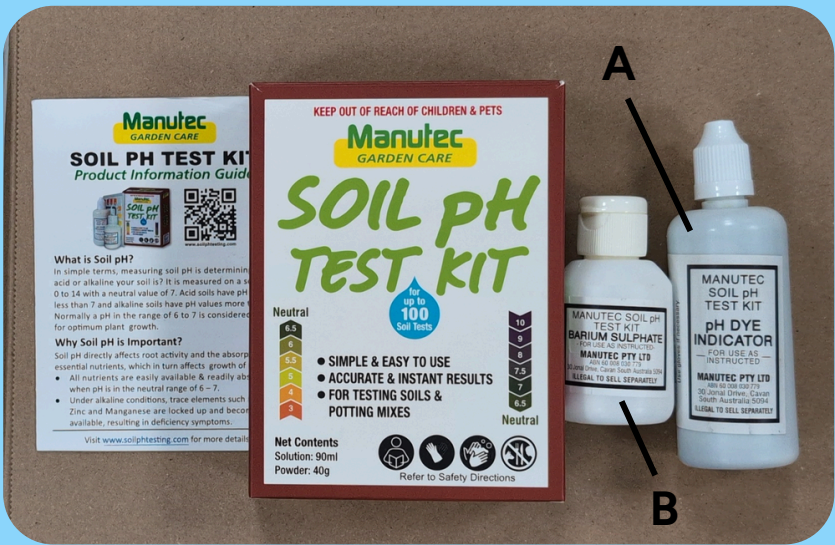


6

Gently wash the sensor with water before measuring the next sample.

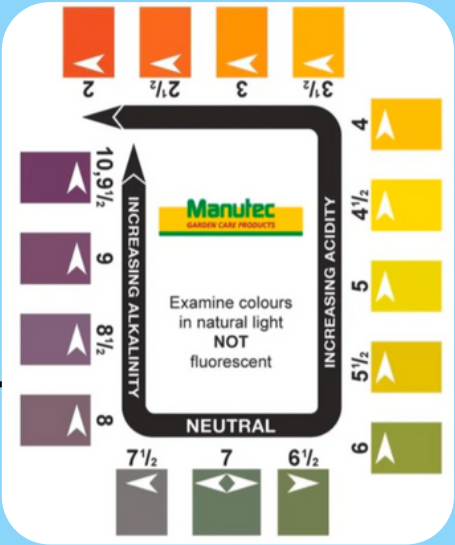


Parts of the Soil pH test Kit



A - Indicator Liquid
B - White powder
(Barium Suphate)

C - pH Indicator card



How to use the Soil pH Test Kit

1

Use the trowel to dig up soil and place a small amount of soil in a SMALL blue capped tube.

Ensure you wash the trowel before collecting another sample.



2

Each tube is labelled with a number (1-7), record the sample (eg. soil 1) and it's location on your Data Collection Sheet.

Sample Type	Location
Soil 1	Garden bed along Royal Park main entrance

3

Place a small amount of soil from each white capped tube into the collection tray.

Make sure the sample number matches the number on the tray.



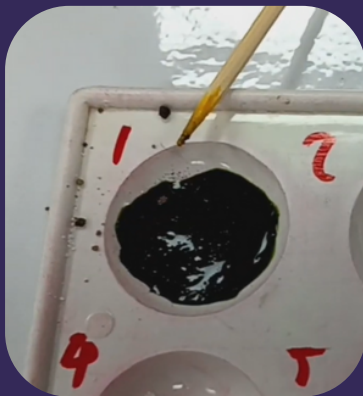
4

Add 3 drops of Indicator Liquid (A) to each soil sample.



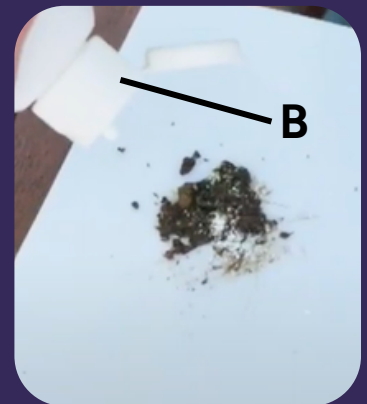
5

Use a toothpick to mix the liquid with the soil.



6

Lightly sparkle white powder from the puffer bottle (B) on to the top of the soil sample - creating a paste.



7

Use the toothpick again to mix. Wait for a minute for the mixture to change in colour.



8

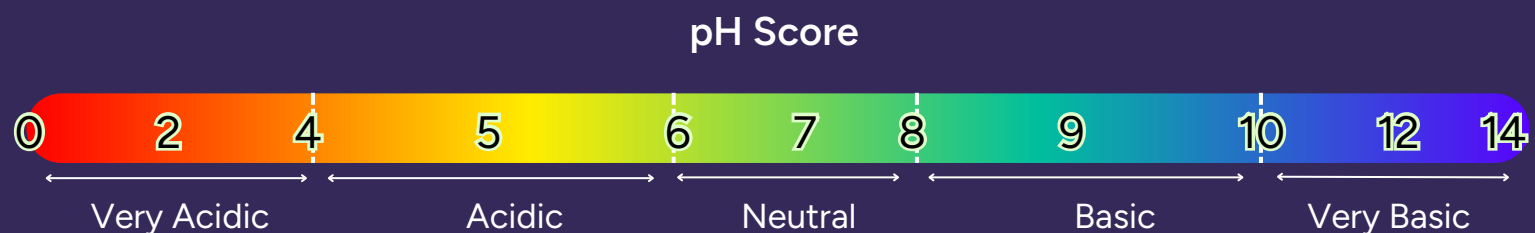
Compare the new colour to the pH Indicator card. Record your result on the Data Collection Sheet.



Understanding the pH score

Understanding your results

The pH score can tell us about the quality of the water/soil. Use the scale below to see if the pH of your samples is acidic, neutral or basic.



- Generally, most plants and animals prefer a pH that is **neutral** (between 6-8).
- An acidic or basic pH is okay for some plants/animals but not ideal.
- A very acidic or basic pH is a problem for most plants/animals, only very specific plants/animals can grow in this type of soil.

What is in the pH Testing Kit?



Water pH Meter



Soil pH indicator



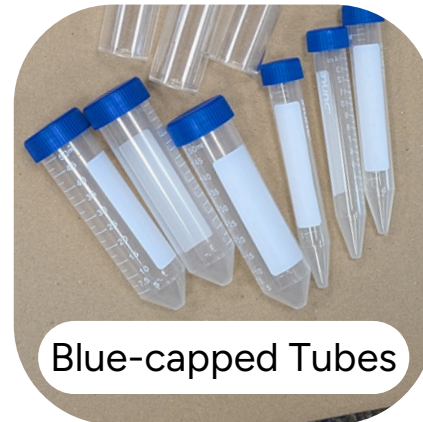
Rechargeable Battery



Collection Tray



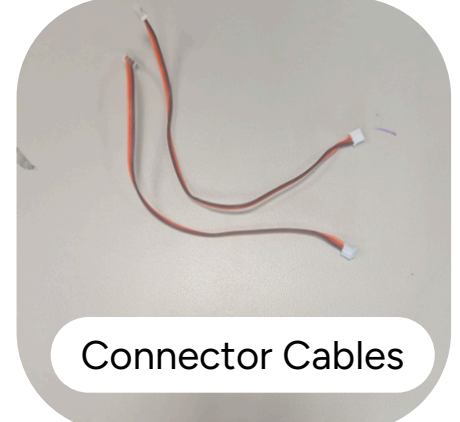
Trowel



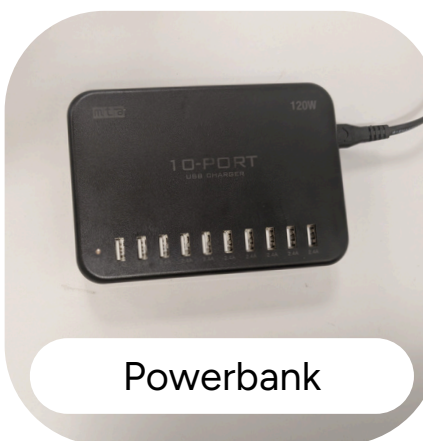
Blue-capped Tubes



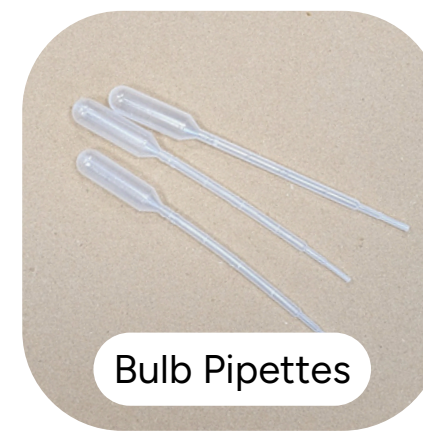
Toothpicks



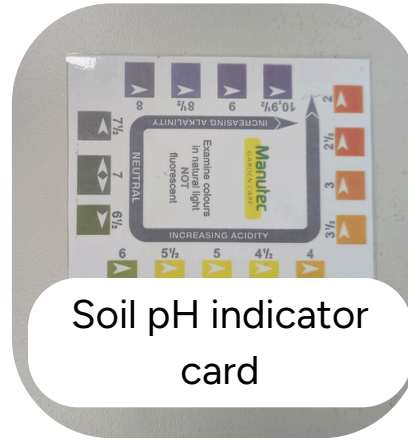
Connector Cables



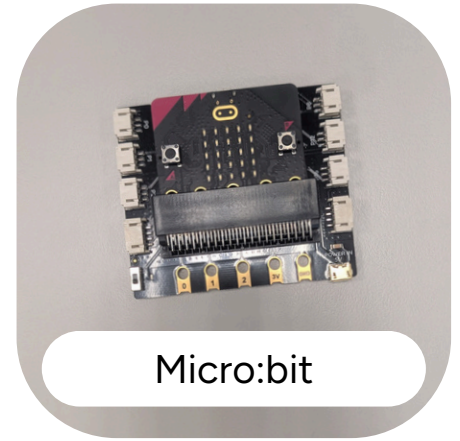
Powerbank



Bulb Pipettes



Soil pH indicator card



Micro:bit

How to analyse the data and reflect on the results

1



Use your results to calculate the Average pH:

- Add the totals on your data table (in the blue and green boxes)

Sample Type	Location	pH Level (1 - 14)	pH Score
Water 1	Royal Park Pond (near the public toilets)	7	Neutral
Soil 1	Garden bed along Royal Park main entrance	5	Acidic
Soil 2	Dirt at the edge of the lake	6	Neutral
	Total Number of Samples: 3	Total Score: 7 + 5 + 6 = 18	

- Calculate the Overall rating using the equation (copy from the blue and green boxes):

Average Score = $\frac{\text{Total Score}}{\text{Total Number of Samples}}$

Average Score = $\frac{18}{3}$

= 6.25

2

What does your result tell you? Think back to your original question (from the 'Ask' Stage)

This result shows that the average pH is within the neutral range of 6-8 meaning the pH is overall safe and healthy for plants and animals to grow and live in.